Wazuh SIEM and Cowrie Honeypot Integration

Objectives

- Deploy a centralized Wazuh SIEM server
- Deploy a Cowrie SSH honeypot
- Forward Cowrie logs to Wazuh
- Monitor attacker activity through the Wazuh dashboard
- Gain hands-on experience with log analysis and security monitoring

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Technologies Used

Technology	Purpose
Proxmox VE	Hypervisor and virtualization platform
Ubuntu Server 24.04 LTS	Operating system for Wazuh and Cowrie
Wazuh 4.13.1	SIEM and security monitoring platform
Cowrie	SSH/Telnet honeypot
Kali Linux	Attack simulation platform
OpenSSH	Remote administration
Linux CLI	System administration and troubleshooting

Lab Architecture

```
Kali Linux
   |
   ▼
Cowrie Honeypot
```

(192.168.1.155)
│
▼
Wazuh Agent
│
▼
Wazuh Manager
(192.168.1.15)
│
▼
Wazuh Dashboard

Part 1: Deploying Wazuh

Virtual Machine Configuration

Setting	Value
OS	Ubuntu Server 24.04 LTS
RAM	6 GB
CPU	2 vCPUs
Storage	80 GB
Network	vmbr0 (VirtIO)

Installation Steps

1. Installed Ubuntu Server 24.04 LTS.
2. Updated system packages.
3. Downloaded and executed the Wazuh installation script.
4. Installed:
   - Wazuh Manager
   - Wazuh Indexer
   - Wazuh Dashboard
5. Verified all services were active.
6. Accessed the dashboard through a web browser.
7. Changed the default administrator password.

Challenges

During installation, the virtual machine initially only used approximately 38 GB of storage despite being assigned an 80 GB virtual disk.

To resolve this:

```
sudo pvs
sudo vgs
sudo lvextend -l +100%FREE -r /dev/ubuntu-vg/ubuntu-lv
```

The filesystem was successfully expanded to the full 76 GB available within the VM.

Part 2: Deploying Cowrie Honeypot

Virtual Machine Configuration

Setting	Value
OS	Ubuntu Server 24.04 LTS
RAM	4 GB
CPU	2 vCPUs
Storage	32 GB
Network	vmbr0 (VirtIO)

Installation Steps

Created a dedicated Cowrie user:

```
sudo adduser --disabled-password cowrie
```

Switched to the Cowrie account:

```
sudo su - cowrie
```

Cloned the repository:

```
git clone https://github.com/cowrie/cowrie.git
```

Created a Python virtual environment:

```
python3 -m venv cowrie-env
source cowrie-env/bin/activate
```

Installed dependencies:

```
pip install --upgrade pip
pip install -e .
```

Created the configuration file:

```
mkdir -p etc
cp src/cowrie/data/etc/cowrie.cfg.dist etc/cowrie.cfg
```

Started Cowrie:

```
bin/cowrie start
```

Verified operation:

```
tail -f var/log/cowrie/cowrie.log
```

Part 3: Simulating Attacks

Using Kali Linux, SSH connections were made to the Cowrie honeypot.

Example:

```
ssh root@192.168.1.155 -p 2222
```

Observed behaviors:

- Failed login attempts
- Successful fake logins
- Command execution attempts
- Username enumeration
- Password guessing activity

Cowrie provided a fake shell environment to attackers while recording all activity.

Example commands tested:

```
whoami
ls
pwd
cat /etc/passwd
```

All commands were logged by Cowrie.

Part 4: Integrating Cowrie with Wazuh

Installing Wazuh Agent on Cowrie

Configured the Wazuh repository and installed the matching agent version:

```
sudo apt install wazuh-agent=4.13.1-1
```

Configured the manager address:

```
<server>
  <address>192.168.1.15</address>
  <port>1514</port>
  <protocol>tcp</protocol>
</server>
```

Started the agent:

```
sudo systemctl enable wazuh-agent
sudo systemctl restart wazuh-agent
```

Configuring Cowrie Log Collection

Edited:

/var/ossec/etc/ossec.conf

Added:

```
<localfile>
  <log_format>json</log_format>
  <location>/home/cowrie/cowrie/var/log/cowrie/cowrie.json</location>
</localfile>
```

Restarted the Wazuh agent:

```
sudo systemctl restart wazuh-agent
```

Part 5: Troubleshooting

Several issues were encountered and resolved:

Agent Version Mismatch

Problem:

Agent version must be lower or equal to manager version

Cause:

- Cowrie VM installed Wazuh Agent 4.14.5
- Wazuh Manager running version 4.13.1

Solution:

```
sudo apt remove wazuh-agent
sudo apt install wazuh-agent=4.13.1-1
```

Invalid Manager Address

Problem:

Invalid server address found: MANAGER_IP

Cause:

Placeholder configuration was not replaced.

Solution:

Updated:

<address>192.168.1.15</address>

Duplicate Agent Registration

Problem:

Duplicate agent name

Cause:

Agent had already been enrolled.

Solution:

Verified enrollment and restarted services.

Results

The integration was successful.

Achievements:

- Wazuh dashboard operational
- Cowrie honeypot deployed
- Wazuh agent connected

- SSH attack activity visible within Wazuh
- Successful log forwarding from Cowrie to Wazuh
- Real-time monitoring of attacker commands and login attempts

The final environment provides a realistic SOC-style monitoring platform capable of detecting and analyzing hostile SSH activity.

Skills Learned

- Security Information and Event Management (SIEM)
- Log Collection and Analysis
- Ubuntu Server Administration
- Linux Troubleshooting
- Virtualization with Proxmox
- SSH and Network Security
- Honeypot Deployment
- Wazuh Agent Management
- Security Monitoring
- Incident Detection and Investigation

Future Enhancements

- Deploy pfSense firewall
- Create isolated security zones
- Build Active Directory environment
- Add Windows 11 endpoints
- Integrate Sysmon with Wazuh
- Create custom detection rules
- Develop security dashboards
- Expose honeypot to the internet for real-world attack collection